Public Transport Demand and Population Growth

A Predictive Modelling Analysis of NSW Opal Trip Data
PRT564 Data Analytics and Visualisation — Assessment 2

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Data: NSW Opal Trip Counts (data.gov.au) + ABS NSW Population Statistics

OVERVIEW

What We Cover Today

01 **The Problem & Our Data**

What TfNSW needs to know and where our data comes from

02 **Data Pipeline & Preprocessing**

ETL, heterogeneous data integration, feature engineering

03 **Exploratory Analysis**

Key visualisations and the surprising inverse finding

04 **Regression Models**

Three models — why we built them and how they work

05 **Statistical Evaluation**

Testing, results, honest limitations

06 **Policy Recommendations**

What this means for Transport for NSW

The Planning Challenge for Transport for NSW

The Challenge

- Sydney is adding 1.3M residents by 2030
- TfNSW must plan capacity BEFORE demand arrives
- But what actually drives public transport demand?
- Is it simply more people — or something else?

Our Approach

- 9 years of Opal tap data
  2016 – 2025, all NSW modes
- NSW Population Statistics
  ABS quarterly data (heterogeneous merge)
- Regression modelling
  3 models of increasing complexity
- Statistical validation
  5 rigorous hypothesis tests

Sources: data.gov.au (Opal NRT Trip Counts) | Australian Bureau of Statistics (ABS 3101051)

ETL Pipeline — From Raw Data to Regression-Ready Dataset

1 Extract
Opal CSV
+ ABS Excel

2 Clean
Standardise modes
Parse dates

3 Integrate
Merge on Year
Population join

4 Engineer
Log transforms
COVID flags

5 Model-ready
102 monthly
observations

Key Features Engineered

Feature	Formula	Purpose
Per_Capita_Trips	Trips / Population × 1000	Normalises demand against growth
Log_Total_Trips	ln(Total_Trips)	Enables elasticity regression
COVID_Flag	1 = Mar 2020 – Dec 2021	Captures 47% demand collapse
Post_COVID_Recovery	1 = Jan 2022 onwards	Captures new lower baseline
Time_Index	0, 1, 2 … 101	Models infrastructure growth trend

The Surprising Finding: Population Growth ≠ More Riders

0.11%
Variance explained
by population alone

−0.41
Population elasticity
(negative direction)

−9.6
Trips/1000 residents
declining per month

3 Periods
Structurally distinct
demand regimes

What the per-capita chart tells us:

01

Pre-COVID (2016–2020)

NSW population grew steadily. But per-capita trips were already declining at −9.6 per 1,000 residents/month.

02

COVID (Mar 2020 – Dec 2021)

A 47% demand collapse — 53.8M → 28.5M monthly trips. This is modelled explicitly with a dummy variable.

03

Post-COVID Recovery

Recovery reached 49.0M/month — still 10.1M below pre-COVID. The old baseline has not returned and may not.

Chart reference: per_capita_trips.png

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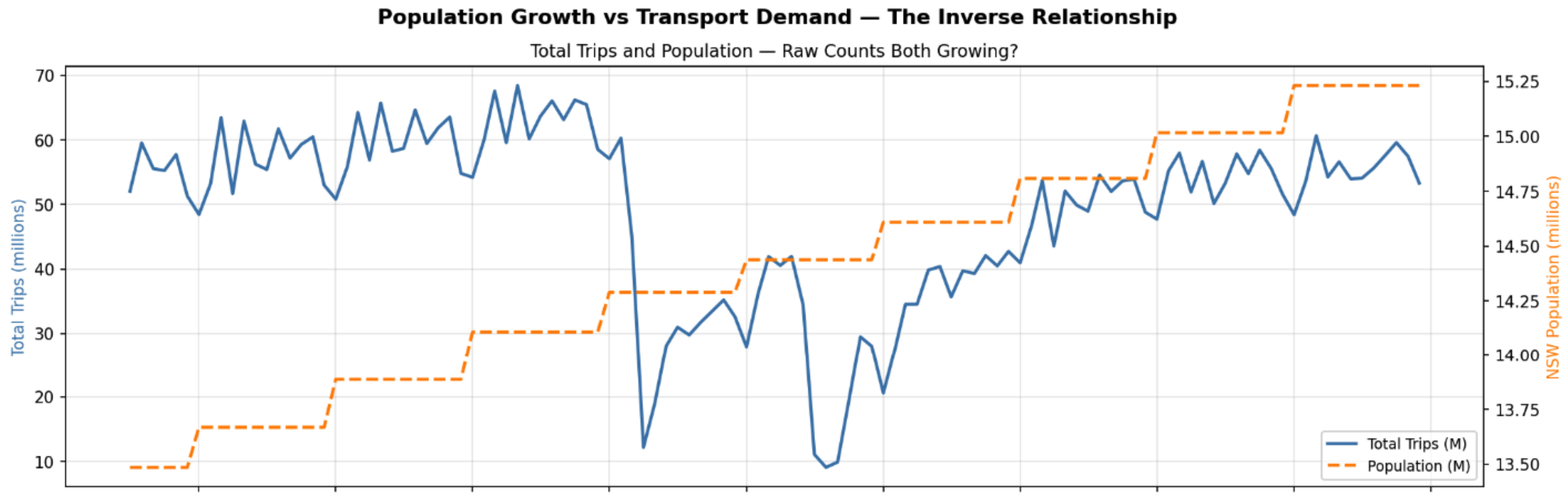

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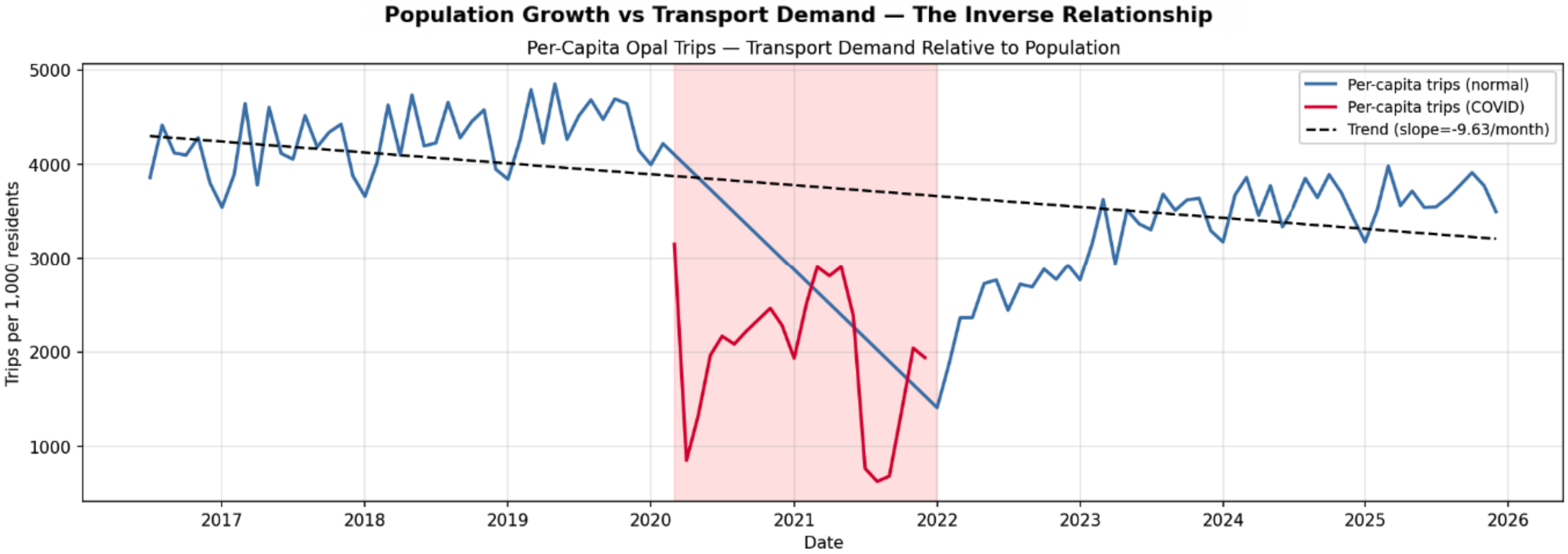

Four Visualisations That Informed Our Modelling Decisions

Seasonal Heatmap

School-term months show consistently higher demand → justifies Month as regression feature

→ See GitHub for chart

Mode Share Over Time

Metro (2019) redistributed mode share — Bus/Train proportions shifted visibly

→ See GitHub for chart

YoY Growth Comparison

Trips grew slower than population in most years → visual proof of demand decoupling

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Correlation Matrix

Log_Population & Time_Index highly correlated → confirms multicollinearity requiring Ridge

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SECTION 03 — EDA VISUALISATIONS

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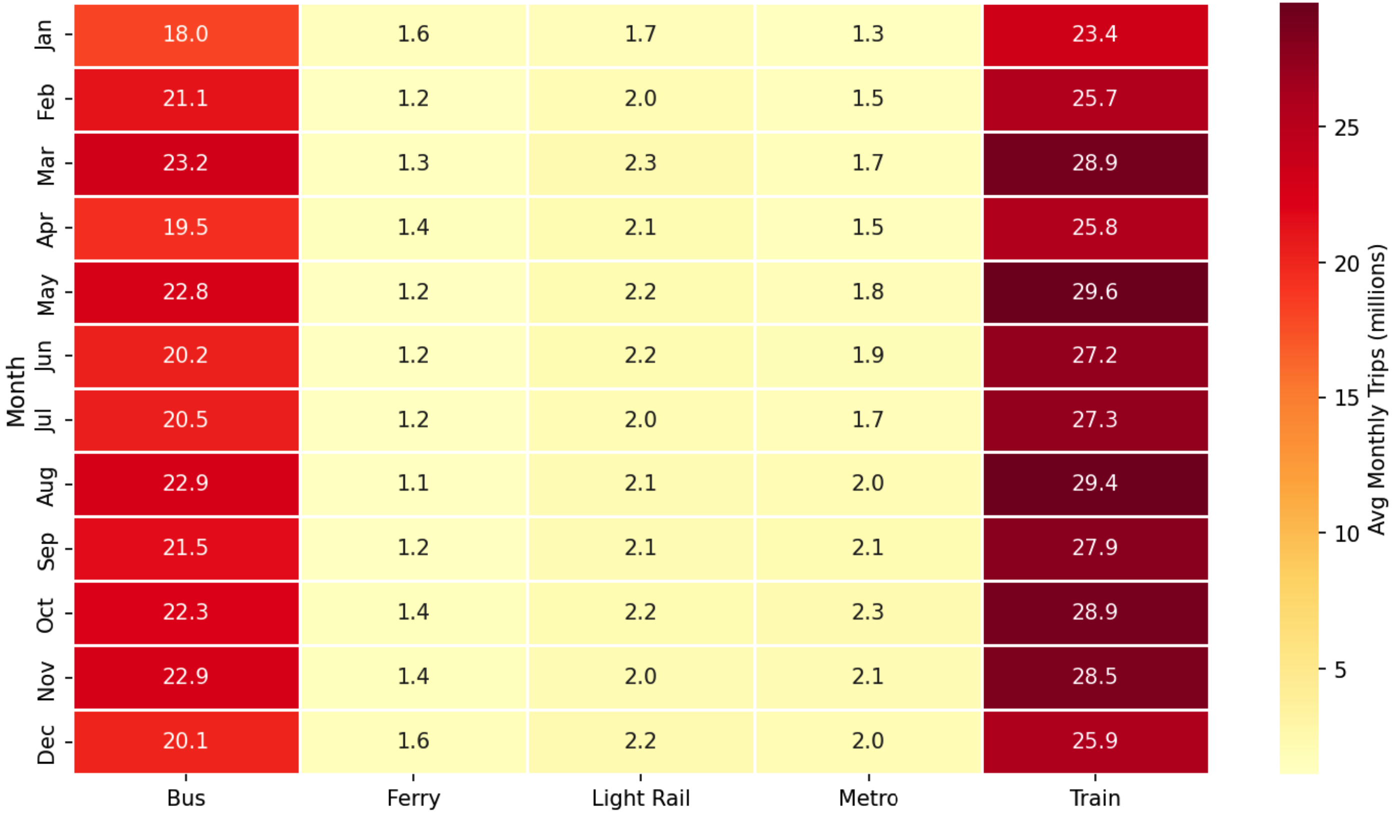

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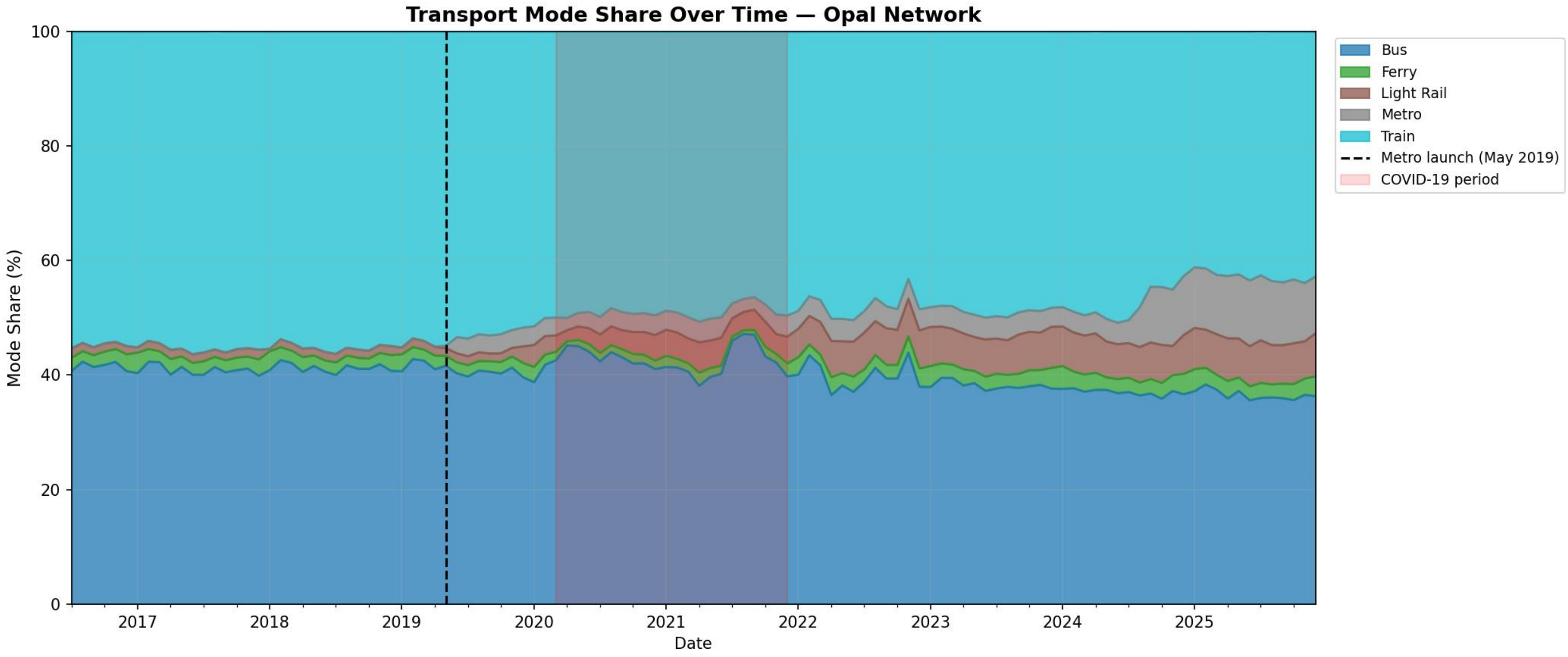

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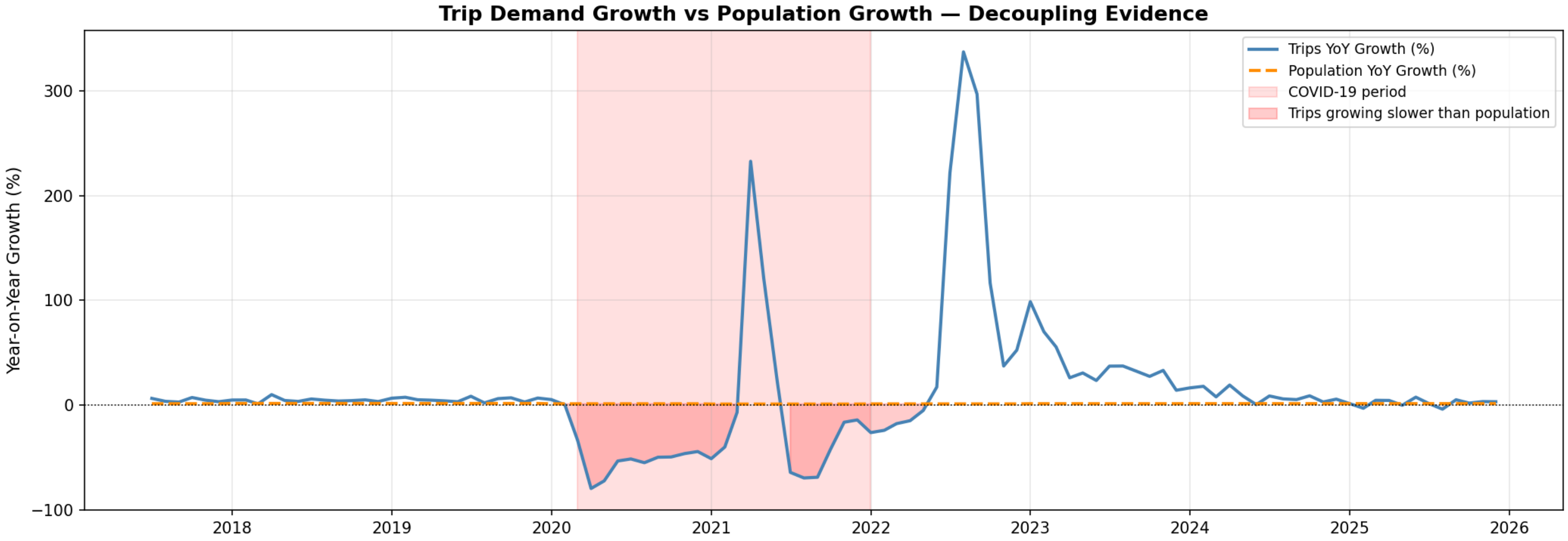

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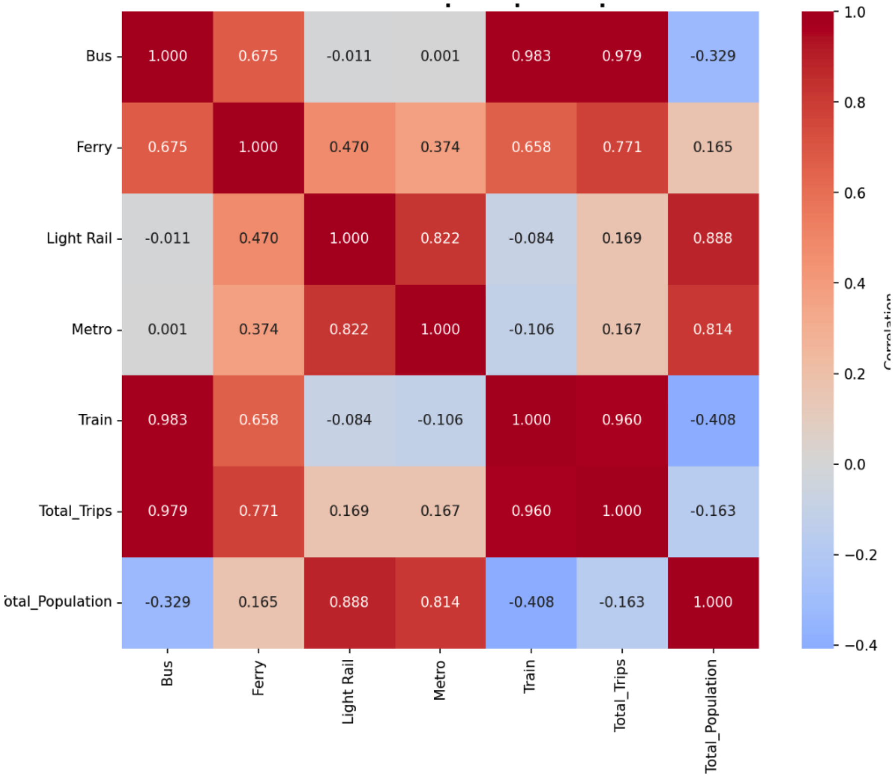


Three Models — A Deliberate Progression

M1

Simple Log-Log OLS

$\ln(\text{Trips}) = \beta_0 + \beta_1 \cdot \ln(\text{Population})$

Why: Baseline: does population alone predict demand?

Result: $R^2 \approx 0\%$ — population explains almost nothing

R^2
0.0011

M2

Multiple OLS + COVID Flag

$\ln(T) = \beta_0 + \beta_1 \cdot \ln(\text{Pop}) + \beta_2 \cdot \text{Time} + \beta_3 \cdot \text{PopGrowth} + \beta_4 \cdot \text{COVID}$

Why: Does adding time trend and COVID structural break improve fit?

Result: $R^2 = 62.7\%$ — massive jump. Infrastructure + shocks matter

R^2
0.627

M3

Ridge Regression (L2, CV-tuned)

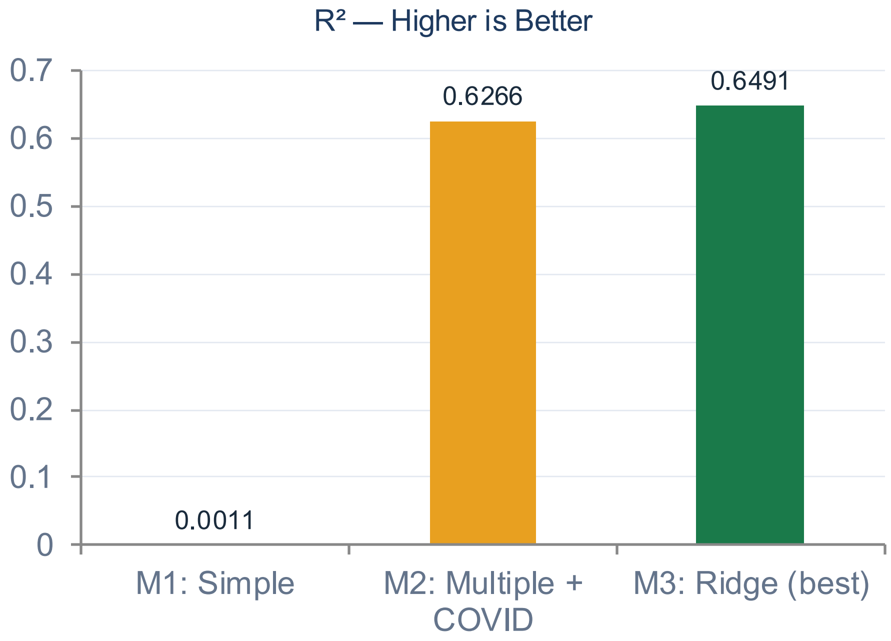
$\ln(T) \sim$ all M2 features + Post_COVID_Recovery + Month $[\lambda=0.005]$

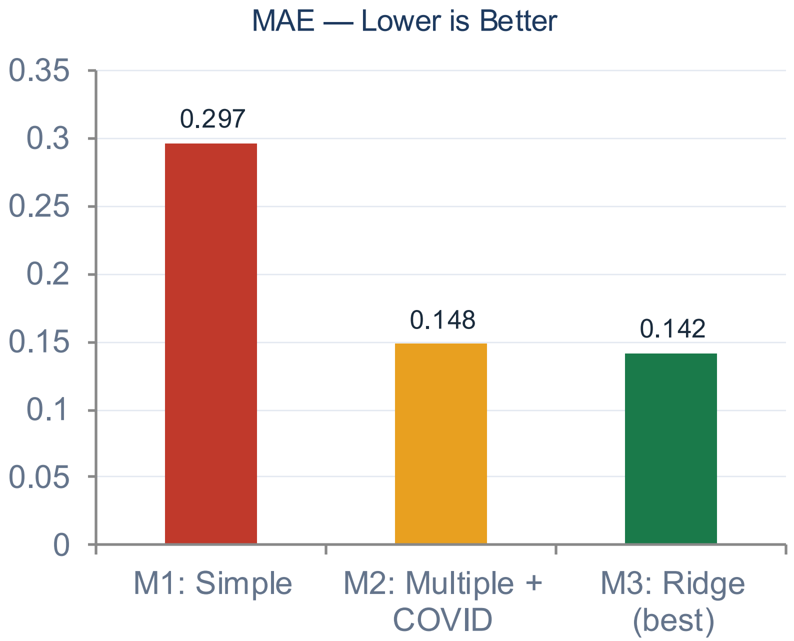
Why: Handles multicollinearity (Log_Pop ↔ Time correlated). Adds seasonality.

Result: $R^2 = 64.9\%$ — best model. λ selected by 5-fold cross-validation

R^2
0.649

Model Comparison — All Three on the Same 102-Observation Dataset





★ Best model: Ridge (M3) — $R^2 = 0.649$, MAE = 0.142. M1→M2 improvement is statistically significant (Wilcoxon $p < 0.001$). M2→M3: directionally positive but $p = 0.12$ — acknowledged honestly.

Five Statistical Tests — Rigorous Validation

T1	**Coefficient t-test (M1)** *H_0: $\beta_1 = 0$ (population has no effect)*	$p < 0.05$ — coefficient is significant	✓
T2	**F-test (M1 overall fit)** *H_0: $R^2 = 0$ (model explains nothing)*	F significant — model structure valid	✓
T3	**Shapiro-Wilk normality** *H_0: Residuals are normally distributed*	All 3 models: NON-NORMAL → use Wilcoxon	!
T4	**Paired t-test (M1 vs M2)** *H_0: No difference in prediction accuracy*	$p = 2.3\times10^{-11}$ — M2 significantly better	✓
T5	**Wilcoxon signed-rank (primary)** *H_0: No difference in prediction errors*	M1 vs M2: significant ✓ | M2 vs M3: p=0.12 (honest)	✓

+ Durbin-Watson: DW = 0.95 on M3 → positive autocorrelation present (expected in monthly data). Limitation acknowledged — ARIMA recommended for forecasting.

What This Means for Transport for NSW

INF

Build infrastructure AHEAD of population

Our Time_Index coefficient shows that service expansion drives demand — not the other way around. Invest before residents arrive, not after.

KPI

Track trips per capita, not total trips

Raw trip counts mask a decline of 9.6 trips per 1,000 residents per month. Per-capita is the honest KPI for network performance.

MTR

Metro investment works — scale it

The 2019 Metro launch is clearly visible in mode share data. New infrastructure shifts behaviour immediately and permanently.

RCV

COVID recovery needs active intervention

10.1M trips/month are still missing vs. pre-COVID baseline. Passive waiting is not enough — targeted service improvements are needed.

Honest Limitations & Future Work

Model Limitations

- $R^2 = 0.65$ — 35% variance unexplained (service quality, fuel prices, WFH rates not modelled)
- Annual population data merged as monthly proxy — assumes stable within-year population
- Positive autocorrelation in residuals (DW=0.95) — standard errors slightly underestimated
- M2→M3 improvement not statistically significant ($p=0.12$) — acknowledged, Ridge still preferred
- No holdout test set — all models evaluated on training data

Future Work

- Use quarterly ABS population data for finer-grained merging
- Add omitted variables: fuel price index, service frequency, congestion data
- Disaggregate by mode — each transport type has different demand elasticity
- Apply ARIMA or Facebook Prophet for pure time-series forecasting
- Implement proper temporal train/test split for unbiased evaluation

Key Takeaways

Population Growth Does Not Automatically Drive Public Transport Demand

0.11%	47%	64.9%	–9.6	10.1M
Variance explained by population alone	Demand drop during COVID	Best model R^2 (Ridge regression)	Per-capita trip decline /1000/month	Monthly trips still missing post-COVID

The answer for TfNSW: invest in service quality ahead of population growth.
Measure success in trips per capita — not absolute counts.

Thank you | Questions?